

Zinc and multi-mineral supplementation should mitigate the pathogenic impact of cadmium exposure.

High-level cadmium (Cd) exposure has long been known to induce nephropathy, severe osteoporosis, and fractures in humans. More recent epidemiology, however, reveals that, in populations not known to have important industrial exposure to this heavy metal, high-normal blood or urine Cd levels correlate with increased risk for vascular disorders, cancers, diabetes, and total mortality, as well as osteoporosis and nephropathy. Since these disorders appear unlikely to expedite Cd absorption, and since Cd has promoted these pathologies in rodent studies, it seems reasonable to conclude that Cd is an important mediating risk factor for these disorders in humans. Avoiding tobacco smoke or frequent ingestion of shellfish or organ meats can lessen humans exposure to Cd, but the chief dietary sources of Cd are plant-derived foods - green leafy vegetables, whole grains, tubers, and root vegetables - typically recommended for their health-supportive properties; indeed, among non-smokers, vegans tend to have the highest Cd body burden. Fortunately, iron sufficiency and ample dietary intakes of calcium, magnesium, and zinc can impede absorption of dietary Cd, both by down-regulating intestinal expression of mineral transporters, and by directly competing with Cd for access to these transporters. Correction of iron deficiency appears to be of particular importance for controlling Cd absorption. Moreover, zinc supplementation can counteract the toxicity of Cd already in the body via induction of metallothionein, which binds Cd avidly via its sulfhydryl groups; so long as it remains sequestered in this form, Cd is innocuous. Zinc supplementation may in any case be recommendable, as optimal zinc status exerts protective anti-inflammatory, antioxidant, and immunosupportive effects. Inasmuch as the toxicity of Cd appears to be mediated in large part by oxidative stress, ingestion of spirulina, lipoic acid, melatonin, and N-acetylcysteine may also have potential for mitigating the risk associated with Cd exposure, as suggested by rodent studies. Hence, although Cd may prove to be a major risk factor for morbidity and mortality in humans, practical strategies for limiting its absorption and pathogenic impact are at hand. Copyright © 2012 Elsevier Ltd. All rights reserved.